

Active inference in Robotics - My PhD at AIRLab Delft

Source: [YouTube Playlist](#): Active Inference & Free Energy Principle ~ Lectures, Podcasts, Discussions **Duration:** 3:00 | **Uploaded:** Unknown **Transcript**
method: auto_caption

[Music] why is it so hard to make robots do stuff and i'm not talking about cute little robots that you know walk around and do simple things but i'm talking about really complex systems and these systems are composed by a lot of subsystems that need to work together and well they need to work together well since the problem is very complex let's try to break it down we can assume that robots are given a set of simple skills such as you know perceiving where an object is through a perception system or navigating from a point a to a point b while avoiding obstacles or some other manipulation skills like how to pick an object or how to place it now how can we combine these skills such that robo can perform something meaningful in the environment my name is corado and i am a phd student at the cognitive robotics department and i'm also part of air lab the ai for retail lab my job is to research on how we can combine the low level skills of such complex systems in order to reliably perform tasks in environments that are designed not for robots but for humans like supermarkets so let's say that the task of the robot is to stock an empty shelf well how should the robot plan and execute its actions to achieve this goal and this might sound simple for a human but i guarantee you that it's very hard in robotics and you should know that it's very easy for robots to fail in environments like this one do we for instance do errors in the perception system or just to unpredictable events that might happen what i'm trying to define in my work is a control architecture to recover from these kind of situations and cope with them so i'm trying to define a robot that has an adaptive behavior and i'm doing so based on active inference a neuroscientific theory which tries to explain how the decision-making happens in our brains the current state of my research we can perform tasks that have some variability but i think we're still far from the level of autonomy that we all dream of however we're working hard to get there in a responsible and a safe way so that we can help our society to advance and well who knows maybe one day one of these guys will actually be part of our daily life you